Rotations

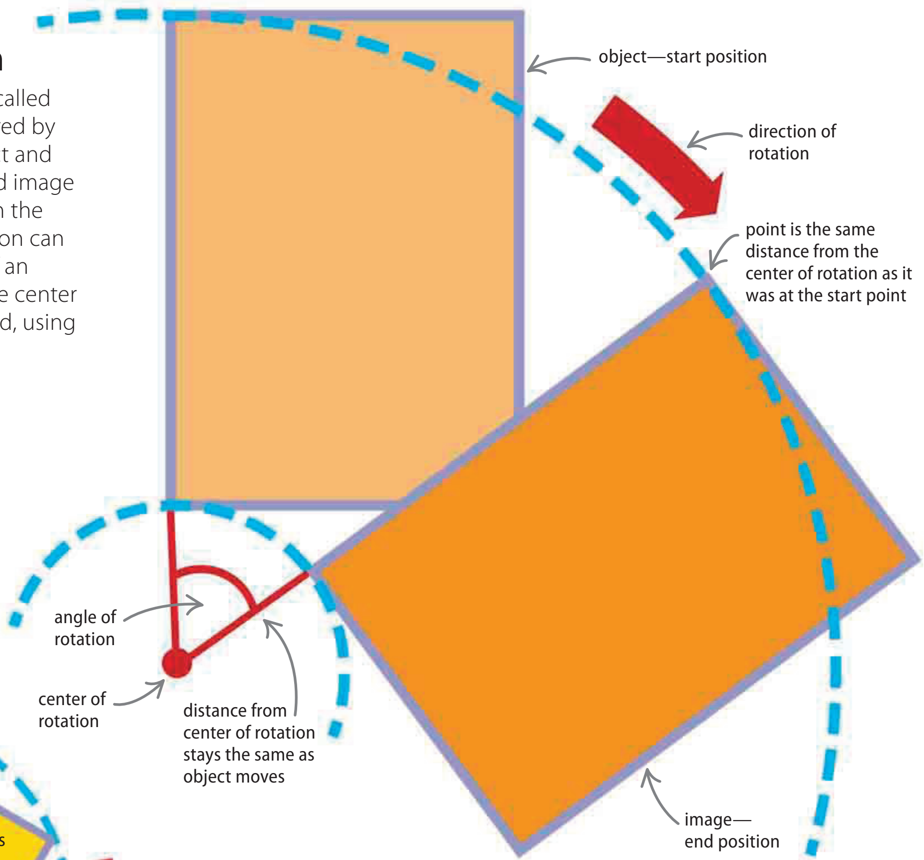
SEE ALSO
❮ 84–85 Angles
❮ 90–93 Coordinates
❮ 98–99 Translations
Reflections 102–103 ❯
Enlargements 104–105 ❯
Constructions 110–113 ❯

A ROTATION IS A TYPE OF TRANSFORMATION THAT MOVES AN OBJECT AROUND A GIVEN POINT.

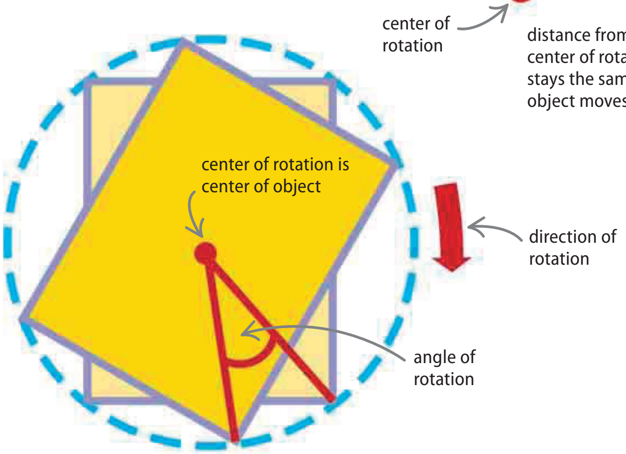
The point around which a rotation occurs is called the center of rotation, and the distance a shape turns is called the angle of rotation.

Properties of a rotation

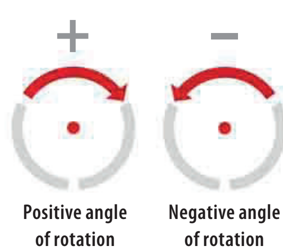
Rotations occur around a fixed point called the center of rotation, and are measured by angles. Any point on an original object and the corresponding point on its rotated image will be exactly the same distance from the center of rotation. The center of rotation can sit inside, outside, or on the outline of an object. A rotation can be drawn, or the center and angle of an existing rotation found, using a compass, ruler, and protractor.



▷ **Rotation around a point**
This rectangle is rotated around a point outside its outline. If it is rotated through 360°, it returns to its starting position.



△ **Rotation around a point inside an object**
An object can be rotated around a point that is inside it rather than outside—this rectangle has been rotated around its center point. It will fit into its outline again if it rotates through180°.



△ **Angle of rotation**
The angle of rotation is either positive or negative. If it is positive, the object rotates in a clockwise direction; if it is negative, it rotates counterclockwise.